

Spatial interdependence and cross-border spillover effects in the ecological footprint of consumption: The role of inclusive policies and public procurement

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ABSTRACT

The post-growth circular economy approach seeks to reorient the economy towards equitable wellbeing enhancement rather than perpetual growth. However, sustainability is not just about environmental protection but also about creating an inclusive and socially equitable society. In this light, the spatial dependencies and spillover effects in the total ecological footprint of consumption are examined across 16 EU countries. The study finds evidence of positive spatial dependence in the total ecological footprint of consumption, indicating spillover effects across neighboring countries while the dual nature of agricultural sector in shaping the footprint is revealed. The temporal footprint dynamics highlight evolving relationships driven by economic activities, governance, and resource management. Employment opportunities contribute not only to the local country's ecological footprint mitigation, but also to an increase in neighboring countries' footprints, highlighting the heterogeneous spatial impacts. Changes in one country's agricultural sector quickly affect neighboring countries, making the environmental impact more widespread. Less transparent procurement practices are linked to higher ecological footprints. A higher share of contracts with no published call for tender creates spillover effects across neighboring countries, while stronger adherence to the rule of law mitigates the ecological footprint and generates positive spillover effects. Sustainable agriculture, regulation of large public contracts, and the adoption of circular economy strategies are vital policy levers to mitigate environmental impacts while accounting for cross-border spillover effects.

1. Introduction

1.1. Problem statement

The idea of circular economy (CE) was initially introduced into the economic-environmental debate by Kathleen, Pearce, and Turner in 1970 (Segerson et al., 1991) and was subsequently proposed as an approach to decrease the ecological footprint during a symposium in Oslo in 1994. Within this framework sustainable patterns were identified as a way to minimize the over-exploitation of natural resources throughout their life cycle (Kowalski and Haberl, 1998). CE calls for a more eco-centric framework that embraces the potential to enhance the resilience of human activities by adopting a natural cycle approach, focusing on reuse and effective management (MacArthur, 2013). It constitutes a significant concept in the transition debate, distinct from other terms like green economies or industrial ecology (Homrich et al., 2018). The CE's role in mitigating the ecological footprint (EF) is to shift

from substituting inputs with externalities, like fossil resources, to reusing production inputs in subsequent cycles (Kowalski and Wiedenhofer, 2014). It aims to maximize input efficiency and minimize externalities. Although there are various interpretations and ongoing debates surrounding the CE's endeavors to minimize EF, there is academic consensus that the extent to which these economies are established depends on a country's ability to provide secondary raw materials through recycling at competitive prices (Arruda et al., 2021). Shifting from a one-way resource flow to a closed-loop materials cycle does not depend solely on circular infrastructure but also on societal behaviors (consumption and production beyond the productive apparatus). The broad range of theoretical underpinnings associated with CE emphasize several CE principles to reduce EF including reduce, reuse, recycle, redesign and remanufacture. Kirchherr et al. (2017) examining 114 CE definitions, note that prior to 2012, the scholars primarily focused on recycling. Specifically, after 2012, greater emphasis was placed on reusing and reducing. However, the new study conducted by Kirchherr

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et al. (2023) identifying 221 definitions, shows that the concepts of degrowth and economic sufficiency are gaining more prominence in CE, and the consideration of 'future generations' has increased significantly, though it remains under-conceptualized. In the study conducted by Nobre and Tavares (2017), 70 CE definitions were analyzed concluding that the central research focus was on reuse, followed by recycling and the extension of product lifespans. By prolonging products' durability and reparability, CE seeks to reduce EF (D'Amato et al., 2019) by minimizing waste and maximizing the value of resources (Ghisellini et al., 2016; Geissdoerfer et al., 2017; Korhonen et al., 2018). Theoretical discussions concerning the anticipated benefits of implementing CE emphasize the spillover effects associated with CE methods (Ghisellini et al., 2016; Stahel, 2016). These discussions highlight that the CE benefits extend beyond consumers and businesses merely adopting and implementing circular methods and have an overall contribution to EF mitigation (Allwood et al., 2011). By doing so, it also promotes economic growth (Busu and Trica, 2019) and job creation. This enables also the recovery and reuse of valuable components, materials, and resources, minimizing the demand for virgin resources (Jawahir and Bradley, 2016; Bianchi et al., 2021). The literature acknowledges the need to address the EF underscoring the significance of collaboration and systemic thinking in successfully implementing CE (Su et al., 2013). The importance of a standardized framework, metrics, and indicators is emphasized to evaluate the CE performance and monitor advancements towards EF mitigation (Kasztelan, 2020; Moraga et al., 2019; Tantau et al., 2018).

However, the academic discussion is marked by a divergence of opinions regarding the CE applicability to reduce EF. One of the main criticisms is that CE may not adequately address the underlying structural and systemic challenges of the current "take-make-waste" pattern (Corvellec et al., 2022; Agrawal et al., 2021). Some critics argue that CE may be insufficient in tackling the fundamental issues of excessive consumption and production due to population and economic growth. They suggest that the dynamics inherent in linear economies, which focus on continuous economic growth, may have limited capacity to adapt to inputs from CE processes (Korhonen et al., 2018). This is precisely the interest of the work on strong circularity, which recognizes the need to take other indicators into account to measure the level of development. Bauwens et al. (2020) identified bottom-up sufficiency, planned circularity, peer-to-peer circularity and circular modernism for the CE future to mitigate EF. By embracing this peer-to-peer circularity approach, the authors argue, societies can make meaningful progress towards EF mitigation through the development and facilitation of product reuse and sharing practices, enabled by the strategic application of technologies and governed by robust institutional frameworks. Although CE holds great promise in terms of reducing EF, the research problem arises due to a still limited understanding of the benefits accompanying the CE (Korhonen et al., 2018) mainly due to the lack of a suitable regulatory and institutional environment (Fernández-Herrero and Duro, 2019; Flachenecker, 2018; Crespi et al., 2016). Without trust, CE initiatives may struggle to gain support from businesses, consumers, and policymakers, making it difficult to create the necessary conditions for a CE transition. Bauwens (2021) stresses that the compatibility of a CE with economic growth is questioned, presenting two potential paths: aligning the CE with growth imperatives or embracing a post-growth economic paradigm. As the various CE business model strategies conflict with the growth imperative in a growth-based economy, Bauwens (2021) considers only the post-growth approach to be viable. The 28th Conference of the Parties on Climate Change (COP28) highlights the importance of addressing the necessary transition away from coal, oil, and natural gas, advocating for a just, orderly, and equitable shift away from fossil fuels. It is noteworthy that these institutional determinants have received relatively limited attention in academic research (Hondroyannis et al., 2023).

1.2. Objectives

Considering the aforementioned criticism of CE and the challenges associated with the EF (Papamichael et al., 2023), the present study sheds light on the spatial dependencies and spillover effects inherent in the total ecological footprint of consumption (EFC) across a sample of 16 EU countries. Specifically, it seeks to examine the presence and magnitude of spatial dependence in the EFC, indicating the extent to which the environmental impact in one country is influenced by its neighboring countries. The heterogeneous impacts of the agricultural sector's employment on the local country's EFC and the spillover effects on neighboring countries are also explored. At the same time, it assesses the impact of large-scale public procurement contracts (those above €130,000) on the EFC, as these large-scale projects can have significant environmental implications through resource use, waste generation, and indirect effects on consumption patterns. The majority of the established research body has primarily focused on analyzing CE at the micro and meso levels, specifically examining consumer and producer behavior. However, the spatial interdependence and heterogeneity in the determinants of the EFC, the role of governance, institutions, and corruption in shaping the EFC, the relationship between the sectoral structure and the EFC, the role of circularity rate, and the dynamic and persistent nature of the EFC have not been adequately explored, to the researcher's awareness. In this light, spatial econometric models will be employed to capture the spatial dependence and spillover effects in EFC while investigating the spatial heterogeneity in the relationships by examining the statistical significance of the spatial autoregressive parameter (ρ). The study includes indicators like the rule of law and environmental policy stringency to evaluate the effectiveness of governance and institutions while considering corruption detection through contract monitoring (e.g. the number of awarded contracts above a certain threshold). The circularity rate (CMU) is examined not only as an independent variable but also as an endogenous variable in VAR models (Table 6) and is recommended by Eurostat and the European Union to measure a country's efforts in waste collection for recovery, which is a vital aspect for the EF mitigation (McDowall et al., 2017; Halkos and Aslanidis, 2023). It is calculated as the amount of waste recycled in domestic recovery plants divided by the total of domestic material consumption and the volume of waste treated. Circularity rate generally $\uparrow$ as the volume of waste processed $\uparrow$ and domestic consumption $\downarrow$, while $\downarrow$ when the volume of waste treated $\downarrow$ and domestic consumption $\uparrow$. This CE concept is also examined by Hondroyannis et al. (2024), aligning with the views of Ghisellini et al. (2016) and Stahel (2016). It is worth noting, that due to the umbrella nature of the CE concept (Homrich et al., 2018) additional indicators related to the environmental CE dimension (Sardianou et al., 2024) include recycling rates, particularly for municipal and packaging waste as well as the share of waste diverted from landfills. Eurostat also monitors waste generation by type and the treatment methods employed, such as recycling and composting. Other metrics assess the economic CE dimension (Sardianou et al., 2024), such as employment in the recycling sector and the value added from CE products and services within various industries.

1.3. Research questions

The present study introduces and sheds light on the presence and strength of spatial dependence in EFC across the EU countries, suggesting spillover effects due to factors like corruption, institutions, and societal behaviors. The heterogeneous impact of employment in the agricultural sector is also revealed, where it is associated with a decrease in the local country's EFC but an increase in neighboring countries. Assessing the role of large-scale public procurement contracts (above €130,000) in increasing the EFC, highlights the environmental implications of resource-intensive large-scale projects. This approach recognizes that creating a sustainable future requires more than just reducing waste and improving resource efficiency; it also requires addressing

broader social and economic inequalities and ensuring that everyone has access to the CE benefits. It quantifies also the short-term effect of CMU on reducing the EFC, and the persistence of past EFC values on current levels. Considering the previous discussion, the research questions could be summarized as follows.

- To what extent are the effectiveness of governance and institutions spatially associated with the EFC?
- What is the relationship between corruption detection mechanisms and their spillover effects on the EFC?
- What is the role of sustainable and social development in shaping the spatial distribution of the EFC?
- How do spatial dependence and spillover effects in the EFC manifest across countries?

The paper's structure is organized as follows: Section 2 provides an overview of the theoretical CE underpinnings regarding the efforts to reduce EF. Section 3 delineates the theoretical framework employed, the data collection procedures, and the model structure. The empirical results are presented in Section 4, while Section 5 offers a comprehensive discussion of the findings. Finally, Section 6 concludes and puts forth recommendations for future research directions.

2. Mitigating ecological footprint: theoretical CE underpinnings

The fundamental focal CE point is to optimize resource value throughout their entire lifecycle while minimizing waste generation (Pomázi and Szabó, 2020; Robaina et al., 2020). It is crucial for driving economic growth and addressing resource scarcity because it allows for the reuse and consumption of resources to mitigate EF (Andrews, 2015; Upadhyay, 2020). Khan et al. (2019) focusing on six countries from Asia, 23 from Europe and five from America during the period 1995–2017, reveal that financial development plays a crucial role in reducing the EF, with its effectiveness varying by region. In all regions, increased use of renewable energy reduces the EF, while higher energy consumption contributes to greater emissions. Examining 24 lower middle-income countries from 1990 to 2015, Khan et al. (2018a,b) identify regional variations in the factors influencing the EF. In Asia, urbanization and GHG emissions influence each other, while energy use and renewable energy impact emissions. In America, financial development and energy consumption affect GHG emissions. In Europe, there is evidence of causality from GHG emissions to financial development and energy consumption. In Africa, trade openness correlates with GHG emissions, and both Asia and Africa exhibit bi-directional causality between forest area and factors such as improved sanitation and financial development. Kazemzadeh et al. (2022a,b) focusing on 98 countries during the period 1990–2014 analyze how increasing economic complexity and the production of advanced goods impact the environment and evaluate whether improving export quality benefits the environment. They conclude that GDP and non-renewable energy use intensify the EF across all levels, though their effects lessen at higher quantiles. Meanwhile, economic complexity exacerbates the EF at lower quantiles but mitigates it at higher ones. Ali et al. (2021), in an empirical study of 128 countries for the period 1995–2019 confirm the Environmental Kuznets Curve across different income levels. In high-income countries, economic growth increase the EF as it does in lower-middle and low-income countries. Through the encouragement of resource reuse, reduction of waste decomposition, and establishment of a recycling-focused industry, CE promotes sustainability (Urbinati et al., 2017; Jakhar et al., 2019). Different types of economic activities and production processes can have varying EF results (Gan et al., 2013). Kazemzadeh et al. (2023a,b) examining 103 countries for 2018 using necessary condition analysis and Fuzzy-set Qualitative Comparative Analysis models conclude that pathways such as low urbanization, weak institutions, fragile states, and high energy use contribute to higher EF. According to Blomsma and Tennant (2020), users commonly disposed of products with regular

household waste, resulting in landfills without recycling efforts. The focus on "product integrity" as advocated by den Hollander et al. (2017) emphasizes the preservation of resources invested in creating goods, thereby eliminating the need for acquiring replacements.

Robaina et al. (2020) argue that countries with a greater proportion of their production level coming from the tertiary sector tend to reduce EF stemming from heightened levels of resource productivity. Khan et al. (2018a,b) focusing on Pakistan during the period 1981–2015 found that factors such as agricultural output, coal-based electricity, renewable energy, hydroelectricity, and forest area, influence the EF. The study identifies unidirectional causality from renewable energy, hydroelectricity, and forest area to reduced emissions while agriculture and forest area exhibit bi-directional causality. Giljum et al. (2010) add another layer of complexity to this relationship by suggesting that the level of industrialization bear significance for determining the EF level. The simultaneous occurrence of ecological deficits and excessive social and individual demands has resulted in economic deficits. Kazemzadeh et al. (2022a,b) examining 25 countries for the period 1970–2016 using a panel quantile regression approach, explore how disparities in income and the sophistication of economic systems shape environmental impact. They find that economic complexity significantly affects the EF across quantiles, with lower complexity increasing it at lower quantiles and higher complexity reducing it at upper quantiles, particularly in developed nations. Economic growth positively influences the EF, with a diminishing effect at higher income levels. Fossil energy consumption amplifies the EF, especially at lower quantiles while income inequality increases the EF at lower quantiles.

Kowalski and Wiedenhofer (2014) noted that, the combination of Europe's low fertility rate and stringent immigration regulations implies that overall resource usage in Europe is expected to decline. However, this would not significantly alleviate global sustainability concerns. Despite Europe's declining resource use, its high standard of living continues to serve as a model for the rest of the world, stimulating increased global resource consumption. Conversely, adopting less strict immigration policies in Europe would result in population growth within the continent, accompanied by a proportional decrease in population elsewhere. However, the CE transition would face challenges, and the consequences of this delay would not be limited to Europe alone. Other regions, such as Latin America, would also experience the delayed implementation of CE practices as Vieira et al. (2018) observed in their study, which focused on 39 municipalities within the Greater Sao Paulo area. Waste generation represents a significant loss of valuable resources which increases the EF while lowering CE levels (Halkos and Aslanidis, 2023). Ghazi Alajmi (2016) and Vieira et al. (2018) found an increase in EF while Kazemzadeh et al. (2022a,b) conclude that population growth consistently exacerbates the EF, though its effect stabilizes at higher quantiles. Ali et al. (2021) found that in high-income, lower-middle and low-income countries urbanization reduces the EF. At the same time, higher employment rates can result in a greater demand for goods and services, which may be associated with higher EF. By employing fuzzy-set qualitative comparative analysis, Silva et al. (2024) found that labor rights play a crucial role in achieving lower EF outcomes. However, Halkos and Petrou (2019) demonstrated that employment significantly shapes the volume of waste generated and EF while Namliş and Komilis (2019) and Chen (2010) noted an inverse relationship between unemployment rates and EF.

The nexus between regulatory quality and different EF indicators is examined by De Römpf and Cramer (2020) highlighting the importance of identifying the suitable regulatory framework to facilitate the efficient implementation of circular actions. It is worth noting that environmental regulations may also increase the operating costs and decrease profit margins which may be attributed to the "cost effect" (Foggia and Beccarello, 2020). Although environmental regulations may raise production costs for enterprises, they can also exert external pressure on enterprises to encourage CE activities (He et al., 2020; Du et al., 2020). Muñoz-Pascual et al. (2019) have tentatively confirmed

that CE activities of enterprises under environmental regulations may lower production costs. Shang et al. (2022) found that stringent environmental regulation policies can effectively enhance CE. For example, the EU's Circular Economy Action Plan (EC, 2020) and related regulations such as the Waste Framework Directive (2008/98/EC), the Packaging and Packaging Waste Directive (94/62/EC), the Industrial emissions directive (2010/75/EC) and Eco-design Directive (2009/125/EC) set clear targets for waste reduction, recycling, and sustainable product design. The Paris Agreement, though focused on climate change, indirectly supports CE practices by encouraging resource efficiency and lower EF. The Basel Convention regulates the transboundary movement of hazardous waste, ensuring environmentally sound recycling and disposal. International agreements like the United Nations Sustainable Development Goals (SDGs), especially SDG 12 (Responsible Consumption and Production) drive countries to adopt CE practices while the Global Plastics Treaty aims to reduce plastic waste globally by applying the CE principles (recycling and sustainable design). Regulatory quality refers to the effectiveness of government regulations in achieving their intended goals, such as protecting the environment and ensuring fair business practices. In contrast, corruption involves using one's authority for personal profit, and it can manifest in various ways such as bribery, embezzlement, and favoritism. Examining the case of Romanian companies, Škrinjaric (2020) found that a highly corrupt environment disrupts the CE implementation while increasing the ecological footprint. Large-scale corruption can have negative consequences on the ecological footprint by distorting the allocation of resources, impairing the established legal structures, and reducing the efficiency of governance systems (Bologna, 2016). In nations where institutional frameworks may be weak, the influence of corruption on impeding the advancement of CE becomes more pronounced. Specifically, corruption can hinder the efficient allocation of resources, diverting benefits away from initiatives that promote sustainability. Prioritizing efforts to address corruption is of utmost importance for developing nations to reduce ecological footprint (Abbas et al., 2021). Taking decisive action against corruption is particularly critical in driving a recovery that prioritizes CE actions such as responsible mineral trade practices with a strong environmental focus. Khan et al. (2019) conclude that trade openness reduces the EF, with the effect flowing from trade to emissions. Ali et al. (2021) stress that in high-income, lower-middle and low-income countries trade increases the EF.

Implementing effective anti-corruption measures plays a vital role in controlling environmental degradation by ensuring the efficient implementation of environmental laws and regulations, as well as upholding good governance and bureaucratic standards (Sheraz et al., 2022). Transparent governance systems, robust regulatory frameworks, international collaboration are essential strategies for mitigating the influence of corruption ensuring that they are aligned with principles of CE thinking, fostering economic progress while simultaneously safeguarding the environment. In their study, Liu and Lyu (2024) investigated the impact of economic corruption and the volume of mineral trade on the sustainable development of 21 emerging nations from 2010 to 2021. Their findings revealed that economic corruption acts as a hindrance to CE levels by diverting valuable resources. The existing research findings in the field bolster the understanding that addressing corruption positively impacts CE conditions (Xu and Hussain, 2023; Mehmood, 2021), whereas higher levels of corruption contribute to higher ecological footprint levels (Hassan et al., 2020; Teng et al., 2021; Sheraz et al., 2022; Wenlong et al., 2023). Goutte et al. (2022) yield meaningful perspectives on the role of governance in combating corruption, highlighting the significance of resilient institutions and transparent frameworks. Building upon this, Dokas et al. (2023) delve into the adverse impact of corrupt activities on economic progress, emphasizing the need for effective administrative structures to reduce ecological footprint. Research conducted by Varvarigos (2023) and Lu and Li (2023) investigates the interconnectedness between corruption, environmental

degradation, and the challenges associated with achieving sustainable development goals. Agarwal (2022) shed light on the significance of effective governance, ethical guidance and anti-corruption measures in promoting sustainability. Haldar and Sethi's (2021) research provides evidence that reducing corruption through enhanced institutions directly mitigate CO₂ emissions through the implementation of legal and regulatory measures. Moreover, it indirectly facilitates a transition to cleaner energy sources by influencing patterns of energy consumption. Kazemzadeh et al.'s (2023) study focusing on 94 countries for 2018 highlights the influence of institutions on carbon emissions concluding that high emission intensity is driven by lower institutional quality. Silva et al. (2024) examining 180 countries for 2018 note that the factors driving strong environmental outcomes differ from those driving weak outcomes and lack symmetry. The researchers found that the absence of corruption and the protection of individual and women's rights significantly contribute to strong environmental performance while high corruption levels are associated with poor environmental performance.

It is worth noting, that herding effect theory suggests that corruption can damage the credibility of local investors collaborating with the government, further impeding the adoption of green consumption practices. Social contract theory emphasizes the interdependence between governments and the public, where corruption erodes government credibility and undermines trust, thereby hindering green consumption. Signal transmission theory sheds light on how environmental corruption sends conflicting signals due to weak enforcement and ineffective governance, leading to a loss of public confidence and discouraging green consumption. Lu and Li (2023) focusing on 31 Chinese provinces during the period 2015–2020, discovered that corruption acts as a barrier to green consumption by obstructing government information disclosure and environmental responsibility. In contexts where environmental corruption prevails, the likelihood of investigations decreases, thus fostering the perpetuation of corrupt practices in neighboring areas. Gokcekus and Suzuki (2014) conducted a study analyzing 20 OECD countries during the period of 2004–2010. Their findings revealed that corrupt practices are associated with higher EFC levels through increased consumerism, resulting in a significant surge of 191% in luxury automobile sales within nations characterized by widespread corruption. Padhan and Bhat (2023) conducted a study focusing on developing economies from 1995 to 2020 revealing that higher levels of corruption mean higher EF and lower levels of CE. According to Nygaard (2023), a limited number of countries exert significant control over the markets governing the supply and demand of scarce mineral resources. In Brazil, informal recycling activities have been linked to issues of corruption and poor governance (Velis, 2018). Bouzguenda et al. (2019) and Trudeau (2018) have previously highlighted the importance of such engagement to promote effective sustainability practices. Romano et al. (2021) found that corruption and public maladministration have a negative impact on the EF. As per Legambiente (2019), the employment of corrupt practices is the foremost method of subverting regulations meant to safeguard the environment and generating unlawful gains. Cesi et al. (2019) found that unauthorized waste dumping increases in the context of corruption, and that greater interaction between waste utilities and bureaucrats involved in control activities leads to an increase in illegal waste disposal through corrupt channels. Gumisiriza and Kugonza (2020) noted that corruption has escalated the costs of waste management for both the municipality and its inhabitants. Agovino et al. (2018) arrived at the conclusion that corruption adversely affects the process of separate waste collection and the performance of waste operators regarding waste management. Miao et al. (2023) suggest that the effectiveness of institutions can be assessed based on their capacity to encourage responsible CE thinking. Meanwhile, Raza et al. (2023) stress that the capability and performance of organizations is critical in determining whether technological innovations are able to mitigate EF. In this light, Shahzadi et al. (2022) focusing on 17 developed and 23 developing countries for the period 1995–2018 employ causality analysis and

pooled mean group regression to explore the impact of R&D spending on GHG emissions. The researchers find that R&D and patent applications generally reduce GHG emissions and the EF. However, in developed countries, R&D expenditures slightly increase emissions. In developing nations, patent applications raise the EF due to financial barriers while R&D investment fosters sustainability.

3. Empirical model Composition

3.1. Theoretical framework

Following Bauwens's (2021) empirical study, it is assumed that society's objective is to reduce EFC by adopting the CE principles in environments with different corruption levels. However, could the CE be successfully reconciled with perpetual economic growth, or is a post-growth approach necessary? What specific policy measures are required to implement a post-growth CE that prioritizes human and non-human wellbeing over profit maximization? The post-growth approach entails reorienting macroeconomic goals through measures like community and cooperative ownership with democratic participation, shorter working hours, and reducing advertising. In essence, the post-growth approach seeks to put material loops and CE strategies in service of overall human and environmental wellbeing, rather than perpetual economic expansion. It requires a fundamental rethinking of the purpose and structure of the economy. A post-growth CE would reorient the overarching goals of the economic system away from perpetual growth and profit maximization, and towards enhancing human and ecological wellbeing in a more balanced, equitable, and regenerative way. This would likely involve a significant reduction in overall production and consumption levels, rather than just making existing processes more efficient. Adopting the theoretical framework developed by Baffoe-Bonnie and Ezeala-Harrison (2023), it is assumed that countries are characterized by different corruption levels (c) where:

EFC_i = total ecological footprint of consumption (EFC) of country i during a given period,

$UNCERTAINTY_i$ = the level of agent uncertainty and non-compliance with societal rules, as well as the quality of contract enforcement, for country i during a given period,

$LENIENCY_i$ = the extent at which regulations concerning the ecological sphere fail to impose explicit or implicit pricing mechanisms on actions that contribute to EFC during a given period,

$CIRCULAR_i$ = the level of CE transition by country i during a given

where consequently can be inferred that

$$P_1^C \text{ LINEAR}_i = P_1^C (EFC_i - \text{LINEAR}_i) = P_1^C [EFC_i (1 - m_i)] \quad (b1)$$

where,

$$m_i = \text{CIRCULAR}_i / EFC_i = \text{CE transition level for each country}$$

Accordingly,

$$\text{LINEAR} = \text{UNCERTAINTY} + \text{LENIENCY} = \text{UNCERTAINTY} + P_1^C m_i EFC_i \quad (c1)$$

The objectives outlined in the 28th COP28 necessitates reducing $UNCERTAINTY_i$ while achieving well-being enhancement (Xu and Hussain, 2023; Mehmood, 2021). Moving from a linear system to a CE is not only about the physical structures, but also about how people behave in society regarding what they consume and produce outside of the main system (Gan et al., 2013). In view of this, it is vital for countries to prioritize the CE transition by placing significant importance on increasing m_i . At the same time, countries aim to achieve wellbeing enhancement while considering the limitations outlined in equations (a1) and (c1). Wellbeing enhancement is an adaptability function associated with the level of certainty:

$$WB = WB(A), WB'(A) > 0, WB''(A) < 0 \quad (a2)$$

$$WB = WB(P_1^C(EFC_i)), WB'(P_1^C(EFC_i)) > 0, WB''(P_1^C(EFC_i)) < 0 \quad (b2)$$

where.

WB = Wellbeing enhancement level of country during a given time period,

A = adaptability level of country during a given time period

By substituting equation (a1) into the wellbeing enhancement function and making minor modifications, we obtain

$$\begin{aligned} WB &= WB [A \{ P_1^C (UNCERTAINTY_i (1 + CIRCULAR_i / UNCERTAINTY_i)) \}, \\ &\text{or} \\ WB &= WB [A \{ P_1^C (UNCERTAINTY_i (1 + m_i EFC_i / UNCERTAINTY_i)) \} \end{aligned} \quad (3)$$

The formulation of this optimization problem is presented in the Lagrangean form:

$$\begin{aligned} \text{Max}_{UNCERTAINTY, m} \quad & WB [A \{ P_1^C (UNCERTAINTY_i (1 + m_i EFC_i / UNCERTAINTY_i)) \}] + \lambda_1 \{ P_1^C (EFC_i - UNCERTAINTY_i - LENIENCY_i) \} + \lambda_2 (CIRCULAR - \text{LINEAR} - P_1^C m_i EFC_i) \end{aligned} \quad (4)$$

time period,

LINEAR_i = the adoption of linear economy practices by country i during a given time period,

The level of ecological footprint within a given time period directly depends on the level of CE transition and the adoption of linear economy practices in countries with different corruption levels. In other words,

$$P_1^C (EFC_i = \text{CIRCULAR}_i + \text{LINEAR}_i), i = 1, 2, \dots, n, \quad (a1)$$

where the λ 's are Lagrangean multipliers.

The first-order systems are expressed as follows:

$$WB'A'[-(m_i EFC_i / UNCERTAINTY_i) + 1 + (m_i EFC_i / UNCERTAINTY_i)] - \lambda_1 = 0 \quad (5a)$$

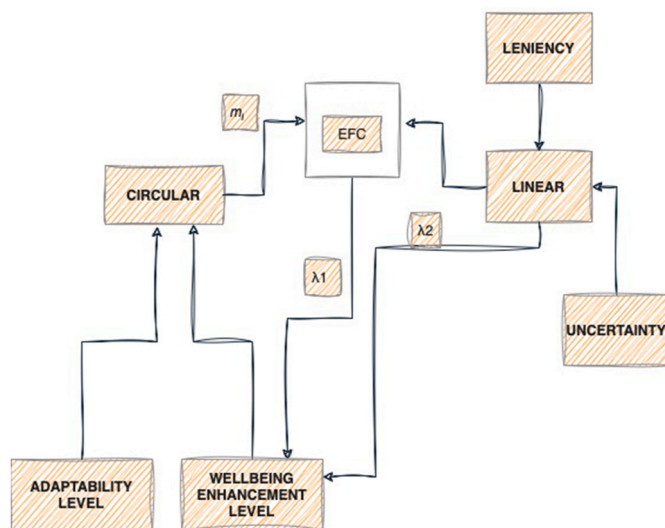


Fig. 1. Theoretical framework.

$$\text{'A'UNCERTAINTY}_i \text{EFC}_i / \text{UNCERTAINTY}_i - \lambda_2 = 0 \quad (5b)$$

$$P_1^C (EFC_i - UNCERTAINTY_i - LENIENCY_i) = 0 \quad (5c)$$

$$\text{CIRCULAR - LINEAR - } m_i \text{ EFC}_i = 0 \quad (5d)$$

Solving equations (5a) and (5b) results in the following:

$$\text{LENIENCY}_i / \text{LENIENCY}'_i \text{EFC}_i = \lambda_1 / \lambda_2$$

It is assumed that $\lambda_1 = \lambda_2$, meaning that, variations in ecological footprint (EFC_i) and the level of linear economy in a country (LINEAR) have equivalent impacts on well-being enhancement. This equivalence arises due to the assumption that any increase (or decrease) in EF contributes to well-being decrease (or increase) in the same manner as the level of institutional environment do (Fig. 1).

Thus,

$$\begin{aligned} & \text{UNCERTAINTY}_i / \text{UNCERTAINTY}'_i \text{EFC}_i = 1, \\ & \text{or} \\ & \text{UNCERTAINTY}_i * = \text{UNCERTAINTY}'_i \text{EFC}_i > 0 \end{aligned} \quad (6)$$

This provides the optimal EFC reduction within each country i during a given time period.

Solving equations (5c) and (5d) results in the following:

CIRCULAR - LINEAR - $m_i \{ P_1^C (UNCERTAINTY_i + LENIENCY_i) \} = 0$,
and substituting (6) we have:

$$\text{CIRCULAR - LINEAR - } m_i \{ P_1^C (\text{UNCERTAINTY}_i \text{EFC}_i + \text{LENIENCY}_i) \} = 0,$$

from which we obtain the optimal CE level for each country i as

$$m_i^* = (\text{CIRCULAR-LINEAR}) / P_1^C (\text{UNCERTAINTY}_i \text{EFC}_i + \text{LINEAR}_i) > 0 \quad (7)$$

This outcome suggests the feasibility of attaining the EFC reduction. Equation (7) signifies that the EFC reduction targets are influenced by factors such as the level of linear economy in a country (LINEAR), the extent to which agents are uncertain and tend not to abide by the rules of society (UNCERTAINTY), the degree to which environmental policies do not put an explicit or implicit price on polluting or environmentally harmful behavior (LENIENCY), the EFC level across countries ($P_i^L EFC_i$) and the overall CE adoption ($P_i^L CIRCULAR_i$).

Based on the developed theoretical framework, the objectives outlined in the 28th COP28 and the previous studies to address this scientific gap, the choice of control variables in the model includes corruption indicators (Lu and Li, 2023; Padhan and Bhat, 2023; Bouzguenda et al., 2019), institutional factors (Halder and Sethi, 2021; Kazemzadeh et al., 2023a,b; Silva et al., 2024), and variables representing the crucial role of CE in mitigating the EFC (Ghisellini et al., 2016; Stahel, 2016; Khan et al., 2018a,b). Corruption indicators reflect inefficiencies and lack of transparency in procurement processes, while institutional factors such as the rule of law and environmental policy stringency capture governance quality. The inclusion of agricultural employment and the circularity rate helps account for the role of sustainable and social development. However, it is worth noting that variables such as the energy sources used in each country could also significantly affect the EFC. For instance, countries relying on fossil fuels would likely have a higher EFC than those using renewable energy. Moreover, public awareness and consumer behavior, such as attitudes toward sustainability and consumption patterns, could also influence the EFC.

3.2. Data collection

The empirical analysis includes 16 EU countries (2010–2019), subject to data availability (extracted on July 03, 2024). These variables include indicators that evaluate the effectiveness of governance and institutions (rule of law, environmental policy stringency index), corruption levels (the number of awarded contracts above 130,000 EUR, the share of contracts with no published call for tender red flag and the

Table 1
Variable representation.

Variable	Category	Definition	Accessible at:	Source	Expected EFC increase or decrease
EFC	Environmental impact	Total Ecological Footprint of Consumption	https://data.footprintnetwork.org	Quality of Government	
AWC	Governance and Regulations	The number of awarded contracts above 130,000 EUR	https://prosperitydata360.worldbank.org/en/dataset/QOG+BD		+
NPC		The share of contracts with no published call for tender red flag	https://datafinder.qog.gu.se/variable/cr_i_nocall		
THC		The share of contracts with tax haven red flag	https://datafinder.qog.gu.se/variable/cr_i_taxhav		
STR		Environmental policy stringency index	https://stats.oecd.org/Index.aspx?DataSetCode=EPS	OECD	–
LAW		Rule of law	https://databank.worldbank.org/source/worldwide-governance-indicators/Series/RL.ES.T	World Bank	
AGR	Sustainable and Social Development	Employment in primary sector as percentage of total employment	https://ec.europa.eu/eurostat/web/main/data/database	Eurostat	–
CMU		Circular material use rate	https://ec.europa.eu/eurostat/web/main/data/database		

Table 2
Summary statistics.

Variable		Mean	Std. dev.	Min	Max
EFC	overall	1.62	0.21	1.09	1.98
	between		0.21	1.22	1.95
	within		0.05	1.40	1.78
STR	overall	1.09	0.17	0.76	1.42
	between		0.17	0.84	1.37
	within		0.04	0.90	1.21
AWC	overall	7.46	1.61	1.10	10.52
	between		1.19	5.00	9.61
	within		1.12	3.56	9.17
NPC	overall	2.50	1.41	-1.96	4.36
	between		1.19	-0.58	3.60
	within		0.80	-0.46	4.35
THC	overall	1.87	0.72	-0.40	3.76
	between		0.47	1.29	3.07
	within		0.56	-0.22	3.41
AGR	overall	1.21	0.61	0.10	2.67
	between		0.59	0.25	2.36
	within		0.23	-0.62	1.69
CMU	overall	2.25	0.61	0.53	3.40
	between		0.61	0.74	3.31
	within		0.16	1.80	2.86
LAW	overall	4.43	0.14	4.10	4.61
	between		0.14	4.16	4.60
	within		0.02	4.34	4.53

share of contracts with tax haven red flag), and employment factors (employment in primary sector as percentage of total employment). The variables, which are publicly accessible, are extracted from the Quality of Governance (QoG) Institute, Eurostat and World Bank reliable databases (Table 1).

3.3. Model structure

Considering the theoretical framework and the corresponding discussion (Vieira et al., 2018; Škrinjaric, 2020; Gardiner and Hajek, 2020), the empirical model is specified as follows:

$$\log(y)_{it} = \alpha + \beta_1 \log(X)_{it} + \beta_2 \log(Z)_{it} + \beta_3 \log(K)_{it} + \varepsilon_{it} \quad (1)$$

where the dependent variable (e.g., EFC i at time t) is denoted by y, and the vector of unknown parameters is represented by β_i . The vector of corruption indicators used in the model (X) includes the number of awarded contracts above 130,000 EUR, the share of contracts with no published call for tender red flag and the share of contracts with tax

Table 3
Cross-sectional dependence tests.

Variable	CD-test
EFC	14.89***
STR	7.45***
AWC	3.39***
NPC	13.46***
THC	2.52***
AGR	10.10***
CMU	6.89***
LAW	1.17
Pesaran's test of Cross-sectional independence	13.23***
Breusch-Pagan Lagrange Multiplier (LM) test of independence	371.297***

haven red flag (Fazekas and Kocsis, 2020). The vector of institutional indicators (Z) includes the rule of law and the environmental policy stringency index (Botta and Kozluk, 2014) with a scale ranging from 0 (least strict) to 6 (most strict). It is measured as the increased implicit or explicit cost for engaging in environmentally harmful activities, encompassing both non-market and market-based approaches to mitigating the EF. The vector of EF reduction strategies (K) includes the employment in primary sector as percentage of total employment (AGR) and the circularity rate (CMU). Table 2 presents the summary statistics of the interest variables employed in this paper.

The results of Pesaran's test indicate a test statistic of 13.23 with a p-value of 0.00. This means that the null hypothesis of cross-sectional independence ($H_0: \rho_{ij} = 0$ for $i \neq j$) is rejected in favor of the alternative hypothesis ($H_1: \rho_{ij} \neq 0$ for some $i \neq j$). This suggests the presence of correlation between the cross-sectional units (i and j) and highlights the need to employ spatial models to account for the observed spatial dependence (Pesaran and Yamagata, 2008). Furthermore, the Breusch-Pagan Lagrange Multiplier (LM) test of independence also reject the null hypothesis of independence. Also, the CD-test statistic showed that for most variables (efc, str, awc, npc, thc, agr, cmu), the p-values are close to 0 meaning that shocks or events in one country can have spillover effects on other countries (Table 3). In this light, more appropriate estimation methods that can handle cross-sectional dependence, such as spatial econometric models, should be considered.

For a cross-sectional data set, a spatial error regression model is formed as follows:

$$y = X\beta + u \quad (2)$$

and

$$u = \lambda W_N u + \varepsilon \quad (3)$$

where $y = (y_1, \dots, y_N)$ is a $N \times 1$ vector and N represents the number of observations. The parameter λ is a 1×1 scalar that captures information about the impact of neighbors on the error term u. The matrix W represents the spatial weights that describe the neighbors of each individual observation. Similarly, X is an $N \times K$ matrix of independent variables, while u and ε are $N \times 1$ vectors of error terms. The matrix W_N is a $N \times N$ spatial weight matrix, where the diagonal elements are equal to 0. The matrix W_N also satisfies certain regularity conditions, such as the requirement that $I_N - \lambda W_N$ must be nonsingular. In this case, the OLS estimator is not efficient as long as the parameter λ is not equal to 0, indicating the presence of spatial dependence. Additionally, a spatial lag regression model is formed as follows:

$$y = \rho W_N y + X\beta + u \quad (4)$$

where $W_N y$ is the spatial lag of y. In this case the OLS estimator is biased as long as $\rho \neq 0$ (3).

The spatial error regression model can be written as:

$$y_t = X_t \beta + \mu + u_t, t = 1, 2, \dots, T \quad (5)$$

and

$$u_t = \lambda W_N u_t + \varepsilon_t \quad (6)$$

where $y_t = (y_{t1}, \dots, y_{tN})'$ is an $N \times 1$ vector of the dependent variable, with N representing the number of cross-sectional units at time t. Similarly, X_t is an $N \times K$ matrix of explanatory variables, while u_t and ε_t are $N \times 1$ vectors of error terms. In this case FE and RE estimators are not efficient as long as $\lambda \neq 0$.

The panel data spatial lag regression model (SAR) is expressed as follows:

$$y_t = \rho W_N y_t + X_t \beta + \mu + u_t, t = 1, 2, \dots, T \quad (7)$$

Based on their contiguity, a 16×16 spatial matrix, which specifies the strength of spatial interactions between neighboring countries, is

Table 4

Panel Unit root and Cointegration tests.

Variable	AGR	AWC	CMU	EFC	LAW	NPC	STR	THC
Null: Unit root (assumes common unit root process)								
Levin, Lin & Chu t*	−447.18***	−813.89***	−265.52***	−975.14***	−436.47***	−240.41***	−226.59***	−795.76***
Null: Unit root (assumes individual unit root process)								
Im, Pesaran and Shin W-stat	−177.35**	−446.38***	0.50	−565.24***	−0.57	−0.73	−175.14***	−380.55***
ADF - Fisher Chi-square	541.04***	792.63***	326.20	926.89***	338.73	440.09*	198.08***	671.21***
PP - Fisher Chi-square	640.26***	909.10***	198.23	996.06***	350.82	356.81	184.08***	574.92***
Kao Residual Cointegration Test		t-Statistic	*10% level of significance. **5% level of significance. ***1% level of significance					
ADF		−4.83***						

generated as follows:

$$W = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In the present analysis, standard unit root tests were performed to determine the order of integration of the series. The results suggest that all variables except the CMU and LAW are integrated of order one. The existence of long-run relationships between the interest variables was explored through a panel cointegration analysis using the Kao test, which provided evidence of cointegration (Table 4). In the next step, Spatial Durbin models (SDMs) with RE and spatial FE, Spatial autoregressive (SAR) models with RE and spatial FE, Spatial error models (SEMs) with RE and spatial FE and Spatial autocorrelation (SAC) with FE, Mixed-effects ML regression, Mixed-effects REML regression ARDL, Vector Autoregression (VAR) and Bayesian VAR models are employed. To ensure the robustness of the spatial econometric models, especially considering spatial autocorrelation and multicollinearity, the models incorporate spatial econometric techniques such as SDMs, SARs, and SEMs, which are designed to capture and account for spatial dependencies among countries. These models explicitly consider the interdependence of observations in neighboring regions, thus mitigating the risk of spatial autocorrelation, which could lead to biased estimates if not properly addressed. To handle multicollinearity, the study assess the variance inflation factors (VIFs) of explanatory variables to detect multicollinearity. The VIFs are low, thus suggesting that multicollinearity is not significant issue in the models and the explanatory variables are not correlated with each other.

Spatial models and mixed-effects ML regression models integrate spatial dependencies and hierarchical structures, crucial for generating reliable forecasts of EFC trends. Spatial econometric models assume that spatial dependency exists, meaning that outcomes in one spatial unit (country) are influenced by outcomes in neighboring units. It is also assumed that the spatial relationships, such as the spillover effects, remain constant over the study period. This may be a limitation arising from spatial interconnections that could evolve due to the dynamic factors such as changing trade patterns or policy shifts. However, unlike non-spatial models, they account for feedback loops and interconnected dynamics wherein changes in one country's EFC propagate over time to influence others, thereby improving prediction robustness. Additionally, mixed-effects ML models are capable of accommodating growing datasets, making them adaptable for analyses beyond the 16 EU countries,

while spatial models ensure scalability across geographically diverse contexts. Specifically, by including RE, the SDM accounts for the spatial spillover effects of the EFC and how consumption patterns in one country can impact neighboring countries. By incorporating RE, the model accounts for unobserved heterogeneity that may influence the EFC but is not captured by the observed indicators. This helps to disentangle the specific contributions of individual countries to the EFC, considering both observed and unobserved factors. The SDM with FE is important for examining the influence of time-invariant country-specific factors on the EFC and how factors unique to each country, such as institutional characteristics and downscaling indicators, affect consumption patterns and the resulting ecological footprint over time. By controlling for these FE, the model helps identify the impact of changes in the indicators of interest on the EFC within each country. The SAR model captures the influence of neighboring countries' consumption patterns on each other while considering both observed and unobserved country-specific factors, allowing for a comprehensive analysis of the spatial interdependencies and similarity in EFC. It is used to examine the extent to which neighboring countries' consumption behaviors are correlated and how this affects the overall EFC. SEM with spatial FE and RE is employed to capture the interplay between corruption indicators, institutional indicators, and downscaling indicators in determining consumption patterns and ecological footprints by assessing the direct and indirect effects of these factors on the ecological footprint while considering the spatial context. Including SAC with spatial FE is crucial for addressing potential spatial autocorrelation in the EFC. Spatial autocorrelation refers to the tendency of nearby countries to exhibit similar EFC. In this light, the model provides more accurate estimates and ensures that the analysis properly captures the spatial patterns and dependencies in EFC across the sample of 16 EU countries.

4. Results

Table 5 presents the results of the models and as can be observed, by taking EFC as a dependent variable, the spatial autoregressive parameter (ρ) capturing the presence and strength of spatial dependence in the data is statistically significant ($p < 0.01$) in all estimated model, indicating the existence of positive [1–6] spatial dependences. A positive value of ρ suggests that if one country has a high (or low) EFC it is likely that its neighboring countries will also have high (or low) EFC. This spillover effect can occur due to the level of corruption, the institutional environment but also about the ways the people behave in society regarding what they consume and produce along with the level of social protection. For example, if a country enacts STR, its EFC will be reduced by lowering consumption of high-carbon goods. However, neighboring countries that don't adopt similar policies will see an increase in their own EFC as businesses shift production to take advantage of looser regulations. This results in a spatial spillover effect where the initial reduction in EFC in the regulated country is counterbalanced by an increase in neighboring countries. Additionally, a country with high levels of corruption probably has weaker environmental regulations and higher EFC. Neighboring countries with more transparent governance will experience higher EFC as well, as consumers in these regions import more goods from the corrupt country that are produced with fewer

Table 5

Regression results. Dependent variable: Total ecological footprint of consumption (EFC).

Variable	a	b	c	d	e	f	g	h
Main Effects								
STR	0.10	0.08	0.07	0.05	0.06	0.03	0.11**	0.01
AWC	0.01**	0.01**	0.01***	0.01***	0.01**	0.01***	0.01**	0.01
NPC	−0.01	−0.01	0.01	0.01	−0.01	−0.01	0.04***	0.04***
THC	−0.04	−0.04	−0.01	−0.01	−0.01	−0.01	0.01***	0.01***
AGR	−0.03**	−0.02	−0.02	−0.01	−0.03*	−0.03	−0.13***	−0.14**
CMU	−0.01	−0.01	−0.02	−0.02	0.00	−0.01	−0.29***	−0.28***
LAW	0.03	−0.10	0.07	−0.05	0.12	−0.01	−0.36**	−0.23
Spatial Spillover Effects								
STR	0.10	0.09						
AWC	−0.02	−0.02						
NPC	0.01*	0.01*						
THC	0.01	0.01						
AGR	0.08***	0.07***						
CMU	−0.03	−0.03						
LAW	−0.30*	−0.22						
Spatial Autocorrelation Parameter								
Spatial rho	0.48***	0.49***	0.52***	0.55***	0.60***	0.59***		
Variance								
lgt_theta	−2.54***		−2.60***		2.86***			

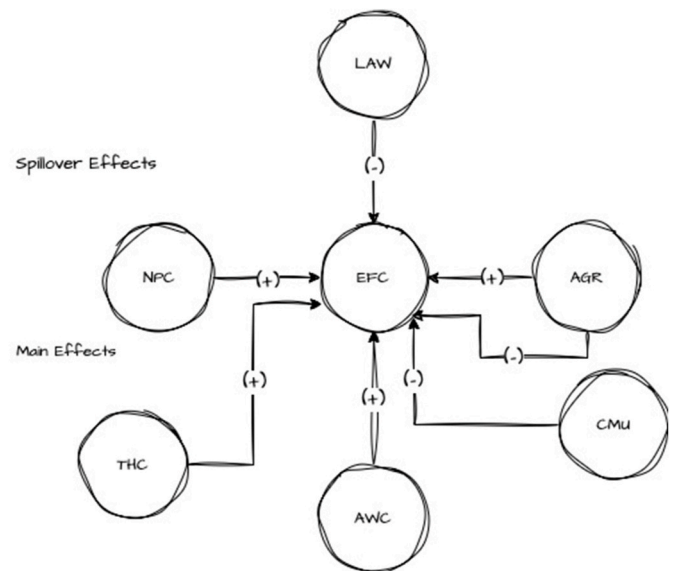
*10% level of significance. **5% level of significance. ***1% level of significance.

^a SDM with RE.^b SDM with spatial FE.^c SAR with RE.^d SAR with spatial FE.^e SEM with RE.^f SEM with spatial FE.^g Mixed-effects ML regression.^h Mixed-effects REML regression.**Table 6**

Robustness Checks with FE.

$EFC_{it} = f(STR_{it}, AWC_{it}, NPC_{it}, THC_{it}, AGR_{it}, CMU_{it}, LAW_{it})$			PCSE standard errors & covariance	White period standard errors & covariance
Variable	Coefficient	P-Values	Robust P-Values	Robust P-Values
STR	0.09	0.40	0.43	0.52
AWC	0.01	0.04	0.04	0.01
NPC	0.01	0.24	0.39	0.23
THC	0.01	0.87	0.88	0.82
AGR	−0.02	0.04	0.05	0.06
CMU	−0.06	0.03	0.01	0.04
LAW	−0.02	0.92	0.93	0.91
C	1.69	0.02	0.02	0.02

environmental controls. Conversely, stronger institutions and anti-corruption measures in one country reduce the EFC in both the country and its neighbors as environmental protection becomes more robust across the region. An important finding is that a 1% increase in the employment in agricultural sector is associated with a 0.03 percentage points (p.p) decrease in the EFC on average, ceteris paribus in the local country compared to the 0.08 increase in the EFC in neighboring regions. The magnitude of the relationship is larger in the spatial model, and the direction of the effect is opposite between the local and neighboring countries implying that the employment in agricultural sector is heterogeneous across space. AGR affects global trade patterns, particularly in countries that rely heavily on agriculture as an export sector. An increase in AGR is associated with increased exports of agricultural products, which in turn affects consumption and production patterns in neighboring countries. The results suggest that there is spatial interdependence, where the EFC in one region is related with the employment factors in both the local and neighboring countries, but in opposite directions. While the local country may benefit from more sustainable agricultural practices, neighboring countries face higher consumption or production-related emissions due to changes in agricultural output or shifts in the regional economy. For example, if a country like Poland

**Fig. 2.** Key findings.

increases AGR in sustainable farming, its own EFC will decrease due to reduced consumption of resource-intensive goods. However, neighboring countries like Germany might experience an EFC increase as demand for agricultural products rises, particularly if the increased AGR means a surge in production that exceeds local consumption and is exported to neighboring countries (Fig. 2). Accordingly, the spillover effects of public procurement practices interact also with sustainable agriculture and a country with high demand for sustainable agricultural products could affect neighboring countries' agricultural sectors adopting more sustainable practices in response to changing market conditions. The EU's Green Public Procurement program encourages governments to adopt criteria that promote environmentally sustainable

practices, including sourcing food from local, organic, or sustainable farms. If a government increases its procurement of locally grown, organic production, it could significantly lower the EFC associated with food consumption, as transportation emissions and resource-intensive farming methods are shrinking. However, if procurement favors large-scale industrial AGR, the environmental benefits of sustainable farming will be undermined, and the EFC will increase.

According to models [1], [3], and [5] and the coefficients of $\lg_{\theta}$, which represents the log of the spatial autoregressive parameter, a spatial heterogeneity is reflected as they are statistically significant ($p < 0.01$), indicating variations in the spatial effects across countries. AWC is estimated positively and statistically significant in all econometric specifications regarding the main effects underscoring that AWC contribute to an increase in the EFC. AWC typically involve larger-scale projects, such as infrastructure development, construction, or procurement of significant goods or services which often require substantial resources, materials, and energy, leading to a higher EFC. Large-scale projects generally require a significant number of resources, such as raw materials, energy, and water. The extraction, processing, and transportation of these resources can have negative environmental consequences, including habitat destruction, air and water pollution, and increased GHG emissions. The magnitude of AWC corresponds also to a higher volume of waste generated during project execution. Construction and industrial activities associated with these contracts produce greater amounts of waste materials, which require proper disposal or management. Larger projects often involve the movement of materials, equipment, and personnel over longer distances and increased transportation contribute to higher energy consumption, carbon emissions, and potential environmental impacts associated with vehicle emissions and infrastructure development. For example, when Germany undertakes a major infrastructure project, it requires the extraction of natural resources that affect the EFC in neighboring countries like Poland or the Czech Republic due to shared water resources (EFC), air pollution, or transportation emissions (EF). Similarly, the construction of high-speed rail networks in Spain increases resource extraction, cement production, and energy use, which not only contributes to Spain's EFC but also affects neighboring France and Portugal. Similar cross-border effects can be observed in the construction of energy infrastructure projects, such as gas pipelines or wind farms, which involve multiple EU countries. AWC has also indirect effects on consumption patterns through their influence on economic activity. According to the model [5] a 1% increase in AGR is linked to a 0.03 p.p decrease in EFC in Europe on average, *ceteris paribus*. Considering the model [7], the positive coefficient on STR suggests that stricter environmental policies are associated with a higher EFC. The result is noteworthy as stricter environmental regulations are expected to reduce EFC, however it could be explained by the fact that more stringent environmental policies drive up the cost of production, meaning higher consumption and a larger EFC. A 1% increase in AWC is linked to a 0.01 p.p increase in EFC, on average, *ceteris paribus*. The positive coefficient on NPC suggests that a higher NPC is associated with a higher EFC. This is concerning, as a lack of transparency in the procurement process can increase the risk of corruption, the inefficient allocation of resources and contributes to higher EFC. A poorly managed infrastructure project in one EU country, where a contract is awarded without proper transparency, results in excessive resource use or improper disposal of waste, negatively affecting neighboring countries' EFC. In some cases, these inefficiencies result in higher consumption of raw materials and energy due to poorly planned projects that lack sustainability considerations, thus increasing the spillover effect to neighboring countries through shared environmental impacts. A higher share of contracts with tax haven red flags is associated with a higher EFC highlighting the potential link between tax avoidance practices and EFC, as companies and individuals engaging in tax haven activities may be less inclined to prioritize environmental sustainability. The results reveal the dual nature of AGR in shaping the EFC. While it reduces local EFC, it also creates

negative spillover effects and higher EFC in the other EU countries meaning that changes in one country's agricultural sector quickly affect neighboring countries, making the environmental impact more widespread. AGR, depending on its scale and practices, either mitigates or exacerbates the EFC. For example, less resource-intensive farming methods, a reliance on local production and consumption, and lower levels of urbanization or industrialization in agricultural regions lower the EFC but large-scale industrial AGR, cross-border resource dependencies, trade in agricultural goods, and shared ecosystems increase it due to water use, chemical runoff and deforestation. A country with high AGR, such as Spain, produces large quantities of resource-intensive crops like tomatoes or olives. The EFC of this production spills over to neighboring countries through water consumption or shared ecosystems, increasing the overall EFC in the EU region. The contribution of the industrial and service sectors to the EFC varies significantly. For example, the industrial sector is typically the most resource-intensive, contributing heavily to EF through high emissions, manufacturing and waste generation. The service sector, on the other hand, while less resource-intensive, still affects the EF through energy consumption in transportation, office operations, and digital infrastructure. A 1% increase in CMU is associated with a 0.28 p.p decrease in EFC [8], on average, *ceteris paribus* emphasizing the importance of reducing waste to mitigate EFC. At the same time, the negative relationship between LAW and EFC [7] suggests that a stronger rule of law means better enforcement of environmental regulations, reduced corruption, more efficient allocation of resources and lower EFC.

To further examine the robustness of the model, additional tests are conducted using the EViews econometric software. Table 6 demonstrates the stability of the coefficient estimates in the FE model by applying alternative coefficient covariance methods, including White Period and Cross-section SUR.

Subsequently, the model's sensitivity was examined to evaluate how variations in model specifications, such as the exclusion of southern or northern countries, influence the outcomes while determining the model's applicability across geographical contexts (Table 7).

According to the model [9], in the short-run period, a 1% increase in CMU, which represents the extent to which resources are being used efficiently, is associated with a 0.22 p.p decrease in EFC in EU countries. Higher CMU reflects more sustainable production and consumption patterns, such as reduced waste generation resulting in a smaller EFC. As circularity practices gain traction, individuals and businesses may become more conscious of their consumption patterns. This awareness can lead to a shift in consumer choices, favoring products and services with lower EFC. According to the model [10], the magnitude of 0.72 and -0.08 of EFC (-1) implies that a 1% increase in EFC the year before is linked to a 0.72 p.p increase in EFC the year after and a 0.08 p.p decrease in CMU on average, *ceteris paribus*. On the contrary, a 1% increase in a country's effort to collect waste for recovery the year before contributes to a 1.31 p.p increase in CMU the year after on average, *ceteris paribus*. The nexus between EFC (-1) and EFC becomes stronger under the Bayesian Var estimation as a 1% increase in the lagged EFC leads to a 0.91% p.p increase in the current EFC emphasizing the

Table 7
Variations in FE model specifications.

Variable	Northern Countries	Southern Countries
	Coefficient	Coefficient
STR	0.09	0.03
AWC	0.01**	0.01**
NPC	0.01	0.01
THC	-0.01	-0.01
AGR	0.04***	0.06***
CMU	-0.02***	-0.17***
LAW	-0.30	-0.44

*10% level of significance. **5% level of significance. ***1% level of significance.

Table 8

ARDL, Vector Autoregression Estimates and Bayesian VAR Estimates. Dependent variable: EFC.

Variable	[9]		[10]		[11]	
	LR	SR	VAR		BAYESIAN VAR	
			EFC	CMU	EFC	CMU
CMU	0.01 (0.02)	−0.22*** (0.09)				
EFC(-1)			0.72 (0.08)	−0.08 (0.14)	0.91 (0.04)	−0.04 (0.07)
EFC(-2)			0.25 (0.08)	0.09 (0.14)	0.05 (0.04)	0.03 (0.07)
CMU(-1)			0.01 (0.05)	1.31 (0.08)	0.01 (0.02)	1.06 (0.04)
CMU(-2)			−0.01 (0.05)	−0.33 (0.08)	0.01 (0.02)	−0.08 (0.04)
R ²			0.94	0.98	0.94	0.98
F-statistic			467.62	1402.00	444.69	1297.60

*10% level of significance. **5% level of significance. ***1% level of significance.

LR = Long-Run Equation.

SR=Short-Run Equation.

Std. Error in parentheses.

persistence and influence of past values on the current levels of EFC and CMU (Table 8).

5. Discussion

EFC provides a way to assess the sustainability of our current consumption levels and production methods. If it exceeds the available biocapacity, which is described by the amount of productive land and resources available, the current lifestyle won't be sustainable in the long-run. CE is widely recognized in the present era as a crucial framework for reducing EFC as it aims to minimize waste and promote the SDGs achievement. The results show strong positive spatial dependence in EFC, indicating that countries with high (or low) EFC tend to have neighboring countries with similar levels. A 1% increase in AGR is correlated with a 0.03 p.p decrease in the local country's EFC but a 0.08 p.p increase in neighboring countries' footprints, demonstrating a heterogeneous spatial relationship. Khan et al. (2018a,b) focusing on Pakistan found that to offset emissions and mitigate the EF, it is of immense importance to promote AGR. The number of large public contracts is also positively associated with increased EFC, likely due to the resource-intensive nature of such projects. In the short-run, a 1% increase in efficient resource use is linked to a 0.22 p.p decrease in EFC. The results align with Silva et al. (2024) who examined 180 countries and found that high corruption is associated with a higher EF. Past ecological footprint strongly influences current levels, with a 1% increase in EFC the year before corresponding to a 0.91 p.p increase the year after. STR is associated with higher EFC, possibly due to increased production costs driving higher consumption. The results complement the studies of Kazemzadeh et al. (2023a,b) focusing on 103 countries finding that reducing EF requires solutions such as enhancing the quality of institutions, rather than just stricter rules. Less transparent public procurement and more tax haven involvement are linked to higher EFC, while greater AGR and stronger LAW are correlated with lower EFC. With increased transparency, EU countries will be able to communicate more effectively about the necessity to eliminate DMC and to call for significant changes in consuming behavior to reduce EFC. More transparent processes drive higher quality of regulations, as institutions and regulations are closely linked. For example, Poland, having one of the largest AGR levels (11.94%), has been criticized for its procurement of agricultural products that rely on industrial farming methods due to lack of transparency. Industrial agriculture relies heavily on chemical fertilizers, pesticides, and intensive land use, all of which contribute to soil degradation, water pollution and higher EFC. Previous research indicates a negative correlation between corruption and various CE indicators as shown by Velis (2018) examining the link between informal recycling activities and corruption, Škrinjaric (2020) investigating Romanian companies and Trudeau (2018) focusing on effective sustainability practices.

Governance and regulations as indicated by STR and LAW have the potential to mitigate income inequality and safeguard vulnerable groups, thereby promoting a greater inclination to support sustainable initiatives, including EFC reductions. They also foster a culture of accountability and promote a sense of responsibility towards others and the environment. For example, in Denmark, strict environmental policies (STR: 3.95/6) are supported by a transparent governance framework (LAW: 97.12), leading to consistent environmental improvements accompanied by a higher CMU (7.6). In contrast, Portugal's lower STR (2.31), marked by weaker enforcement of laws (LAW:84.13) and a higher share of corruption (e.g., NPC and THC), is associated with a lower CMU (2.30) and greater waste. Higher LAW boosts confidence in the business environment, creating a higher level of trust and making it more likely for businesses to invest in CE (Muñoz-Pascual et al., 2019). However, the results show [7] that STR is positively correlated with the EFC probably due to increased production costs, driving higher consumption. While higher LAW and transparency foster accountability and is negatively correlated with the EFC over time [1,7], STR increases EFC as industries adapt to compliance. In the case of agricultural policies, local reductions in EFC can be offset by spillover effects from neighboring regions, particularly when trade and production intensify. Moreover, industrial agricultural methods, heavily reliant on synthetic inputs contribute to higher EFC, despite regulatory efforts. While STR is associated with increased CE levels, short-term economic and spatial factors might counteract these benefits. The relationship between AGR and EFC is heterogeneous across space, with opposing effects in the local country and neighboring regions. Increased AGR in a country means more sustainable and efficient use of local resources, reducing the EFC within that country. However, this decrease in the local footprint can be offset by a spillover effect, where increased agricultural productivity and output in the local country is associated with greater exports and consumption in neighboring regions, causing their EFC to rise. The agricultural sectors, production methods, and trade relationships vary significantly between a country and its neighbors, so what may be beneficial for reducing the EFC locally may not translate the same way in neighboring regions due to differences in factors like technology, policies, and economic structure. Agricultural products and services often cross national borders as part of regional or global supply chains, so an increase in AGR in one country can drive up production and resource use, which then gets incorporated into the EFC of trading partners and neighboring economies. It is worth noting that AGR in Europe significantly shapes the EFC. AGR in intensive farming, reliant on synthetic fertilizers and monoculture practices is associated with higher resource use, greenhouse gas emissions, and soil degradation. On the other hand, organic farming, while reducing chemical inputs, requires more land for equivalent yields. Agroforestry integrates trees into agricultural landscapes, improving biodiversity, soil health, and carbon sequestration, though it increases land use. AGR in regenerative agriculture focuses on

soil health and carbon capture, reduces GHG emissions and water consumption, but requires also more land compared to conventional methods. Livestock farming, particularly beef and dairy, contributes to higher EF through methane emissions and high-water consumption, with intensive systems having a larger EF. On the contrary, precision agriculture uses technology to optimize resource use and is increasingly being adopted by farmers (specifically in Germany) to reduce the use of fertilizers and pesticides. Improved agricultural efficiency and reduced local EFC from increased AGR may also free up resources that then get redirected towards other, potentially more resource-intensive economic activities in neighboring regions, leading to an indirect rebound effect that can offset the initial localized benefits. The inter-relationship between employment and CE has been studied by [Halkos and Petrou \(2019\)](#) concluding that employment exerts a significant effect on waste generation and CE.

6. Conclusions, policy recommendations and further research

6.1. Conclusions

The present study investigates the spatial dependencies and spillover effects in the EFC across 16 EU countries, with a focus on understanding the factors associated with EFC, including employment in the agricultural sector, procurement practices, environmental policies, and institutional factors. The study found evidence of positive spatial dependence and spillover effects in EFC indicating that high or low EFC in one country is likely to be reflected in its neighboring countries. Factors such as employment in the agricultural sector, public procurement processes, circularity rate, and past EFC values were found to be significantly correlated with the EFC, with varying magnitudes and directions across local and neighboring countries. The study also highlighted the potential role of institutional factors, such as the stringency of environmental policies, transparency in procurement, and the rule of law, in shaping the EFC. The findings imply a significant need for cross-border policy coordination to address spatial spillover effects on the EFC. To mitigate these cross-border spillover effects, coordination strategies could include harmonizing environmental standards and procurement policies across EU countries. This involves creating a common framework for environmental policy stringency, setting up joint regulatory bodies to enforce sustainability criteria in public procurement, and encouraging cross-border collaboration on CE practices. Additionally, aligning tax regulations, particularly regarding THC, reduces corruption and environmental loopholes that undermine sustainability efforts. A key strategy could be establishing bilateral or regional partnerships, facilitating the sharing of best practices for sustainable agriculture and CE policies, and providing cross-border incentives for clean technology innovations. Enhancing transparency in procurement through shared platforms helps tackle issues like NPC, ensuring that efficient resource use is prioritized at the regional level.

The temporal EFC dynamics reveal evolving relationships driven by economic activities, governance, and resource management. AWC initially raises EFC due to the resource-intensive nature of large-scale projects. Over time, the long-term effects depend on whether sustainable procurement and green building practices are adopted. High NPC, indicating reduced transparency in procurement processes, contributes to inefficiency and wasteful resource allocation, exacerbating EFC over time. Higher THC, linked to tax haven activities, undermines environmental accountability, creating long-lasting impacts by prioritizing cost savings over ecological sustainability. STR increases EFC in the short-run as industries adapt to higher compliance costs. However, it can significantly reduce EFC in the long-run by fostering systemic changes, such as innovation in cleaner technologies. A stronger LAW reduces EFC consistently over time by enhancing enforcement, reducing corruption, and ensuring efficient resource use. In the agricultural sector, higher AGR initially lowers EFC locally but creates negative spillovers for neighboring countries due to cross-border dependencies and shared

ecosystems. Enhanced CMU, reflecting CE practices, sustainably decreases EFC over time by promoting efficient resource use, waste reduction, and recovery. These dynamics underscore the importance of policy interventions tailored to specific variables to mitigate long-term EFC growth.

6.2. Policy recommendations

The results of the present study emphasize that given the global nature of ecological impacts, greater cooperation and coordination between countries is of paramount importance. Greater regional cooperation is facilitated through EU-wide agreements that establish uniform standards for CE practices and align environmental policies across borders. In this light, shared monitoring frameworks should be created, utilizing centralized databases and tools to track EFC and CE-related progress, thereby enabling countries to learn from and support one another. This could involve establishing international agreements, treaties, or organizations to align policies, and develop joint strategies. The importance of transparency and good governance in public procurement and taxation systems is also stressed, as a lack of transparency increases the risk of corruption and inefficient resource allocation, contributing to higher EFC. The results show that AWC and THC increase the EFC, while NPC increases the EFC and creates spillover effects. It underscores the need for policymakers to mandate the publication of all public procurement contracts above a specific threshold through centralized and accessible digital platforms. Real-time updates on procurement processes are required, from calls for tender to contract awards, similar to European Single Procurement Document initiative, ensuring comprehensive visibility for stakeholders. An EU Procurement Integrity Network should also be developed to disseminate strategies for reducing corruption and mitigating its EFC impacts. It reinforces the significance of encouraging collaboration among EU countries to align procurement laws and share best practices for tackling red flags. It is also of paramount importance to establish independent oversight bodies to monitor and investigate corruption cases related to CE initiatives, while developing and enforcing stringent penalties for corruption and fraudulent practices. To improve transparency and governance in public contracts, an EU-wide e-procurement platform should be created that integrates sustainability metrics for awarded contracts while penalizing companies or public bodies that fail to meet minimum sustainability criteria. In this light, mandatory independent audits should be implemented for high-value contracts and whistleblower protection laws should be introduced. A unified EU-wide database of procurement contracts should also be developed to monitor trends and identify irregularities, particularly for cross-border contracts. Fines proportional to the contract value should be implemented while blacklisting companies repeatedly flagged for irregularities, following the World Bank's Sanctions System. To address the negative spillover effects of increased AGR on neighboring countries' EFC, strategies should include regional agricultural collaboration for sustainable practices, promoting precision farming and regenerative agriculture, and incentivizing CE initiatives. Eco-labeling targeted environmental taxation, and cross-border environmental monitoring further align agricultural practices with EFC mitigation. To encourage sustainable agriculture practices, strict guidelines on agricultural resource usage (water, pesticides, fertilizers) should be set and subsidies to the adoption of sustainable farming practices should be linked. Funding for R&D of new, eco-friendly farming technologies should also be increased. Green building standards and CE principles should be adopted in manufacturing and construction while conscious consumption focused on durable goods and waste reduction should be encouraged, supported by robust regulatory frameworks. The CE transition will also be facilitated when policymakers ensure political stability by fostering a favorable business environment and providing policy stability for CE investments. Simplifying regulatory procedures while reducing bureaucratic barriers will make it easier for businesses to comply with

EFC reduction, as demonstrated by the negative relationship between LAW and EFC. At the same time, policymakers should focus on developing more effective, well-designed, and enforceable environmental policies that drive genuine environmental improvements, rather than just increased compliance costs as indicated by the positive relationship between STR and EFC. It is worth noting that, the CE benefits must be shared by all members of society, which can be achieved through equal employment opportunities, low corruption, and high regulatory quality. To shape this "circular economy of belonging", policies and practices that prioritize social inclusion and equity are of immense importance, alongside environmental sustainability. However, caution is important when interpreting the empirical results of our investigation, as factors such as data availability must be considered. The study's focus on the EU suggests that the results are most relevant to countries with similar economic, institutional, and regulatory contexts, such as other developed economies or those with similar governance systems. However, for non-EU countries or regions with significantly different industrial bases, agricultural practices, and environmental policies, the generalizability of the findings could be limited. In countries with weaker governance structures, the implementation of policy recommendations derived from the study's findings presents several challenges. For example, countries with less robust governance systems struggle to enforce transparency in procurement, curbing corruption, and ensuring the efficient allocation of resources. At the same time, resistance from stakeholders, especially those benefiting from the status quo, hinders policy changes. In countries with lower LAW levels, implementing STR could face significant opposition.

6.3. Further research

While the study highlights the presence of spatial spillover effects, the specific mechanisms driving these effects are difficult to isolate. For example, while AGR is linked to a decrease in the local country's EFC but an increase in neighboring countries, the underlying drivers of these effects (such as cross-border trade, shared environmental resources, or migration) are not fully unpacked. The study assumes that these spillovers are due to governance and institutions factors (rule of law, environmental policy stringency index), corruption levels (the number of awarded contracts above 130,000 EUR, the share of contracts with no published call for tender red flag and the share of contracts with tax haven red flag), and employment factors (employment in primary sector as percentage of total employment) but other spillover channels (economic, social, political) may also play a significant role in shaping the EFC. The estimated models assume also homogeneous effects across all countries, but the strength and nature of spatial spillover effects might differ significantly due to the diversity of eco-institutional contexts within the EU. For instance, the impact of AGR on EFC may be stronger in countries like Poland, where AGR is a significant part of the economy (11.94%), compared to countries like Belgium, where AGR is much smaller (1.37%).

Future research should investigate sector-specific drivers of spillovers, such as the cross-border EFC of agricultural exports or industrial supply chains, to assess the role of manufacturing hubs or shared agricultural practices in shaping EFC patterns. Additionally, research on technological advancements, like renewable energy adoption or precision agriculture, could clarify their mitigating effects on EFC. To further explore the "circular economy of belonging" it is essential to delve into the role of inclusivity for a more equitable CE. In this light, the role of grassroots or community-driven CE initiatives and how they foster inclusivity should be investigated. This includes studying local exchange systems and cooperatives to understand how they create socio-economic opportunities for marginalized groups. Case studies in countries with distinct eco-institutional contexts, such as Poland's reliance on agriculture versus Belgium's industrial focus, would help identify country-specific drivers of EFC and refine spatial econometric models. Future research could also assess the impact of financial incentives, such as

green subsidies or penalties for high-EFC activities on spatial spillovers. Comparative analyses of countries with varying levels of such incentives could offer lessons on their effectiveness. Lastly, to further examine the role of institutional effectiveness on EFC mitigation, case studies should be conducted to evaluate the direct effect of governance reforms, such as anti-corruption measures in procurement on reducing EFC spillovers.

Consent for publication and competing interest

The study represents original material that has not been published elsewhere. The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration

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Data availability

Data will be made available on request.

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